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Lysosomal Dysfunction at the Centre of Parkinson's Disease and Frontotemporal Dementia/Amyotrophic Lateral Sclerosis

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Abstract

Parkinson's disease (PD) and frontotemporal dementia/amyotrophic lateral sclerosis (FTD/ALS) are insidious and incurable neurodegenerative diseases that represent a significant burden to affected individuals, caregivers, and an ageing population. Both PD and FTD/ALS are defined at post mortem by the presence of protein aggregates and the loss of specific subsets of neurons. We examine here the crucial role of lysosome dysfunction in these diseases and discuss recent evidence for converging mechanisms. This review draws upon multiple lines of evidence from genetic studies, human tissue, induced pluripotent stem cells (iPSCs), and animal models to argue that lysosomal failure is a primary mechanism of disease, rather than merely reflecting association with protein aggregate end-points. This review provides compelling rationale for targeting lysosomes in future therapeutics for both PD and FTD/ALS.

Do Lysosomes Have a Central Role in PD and FTD/ALS?

PD is the second most common neurodegenerative disorder, affecting 1–2% of the population over the age of 65. The cardinal symptoms of PD result from the loss of dopaminergic neurons in the substantia nigra pars compacta (SNc; see Glossary), causing a loss of dopamine (DA) in the striatum. FTD and ALS are adult-onset neurodegenerative diseases that are closely related to each other both pathologically and genetically. Clinically, these two diseases appear distinct: FTD primarily affects cortical neurons in the frontal and temporal lobes, leading to various forms of dementia, language aphasia, and behavioural changes, whereas ALS is a disease of motor neurons, resulting in the gradual loss of voluntary muscle movement. These diseases have traditionally been defined by postmortem

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endpoints of specific neuronal loss and protein aggregates positive for α -synuclein (SCNA) for PD, and Tar DNA-binding protein 43 (TDP-43)/Tau/Fused in sarcoma (FUS) and/or ubiquitin for FTD/ALS. However, the discovery during the early 2010s that mutations in the gene encoding Chromosome 9 open reading frame 72 (*C9ORF72*) can cause both FTD and ALS [1,2] highlighted the underlying converging pathophysiology in FTD/ALS, at least in some of forms of these diseases.

Lysosomes are cytoplasmic membrane-enclosed organelles containing hydrolytic enzymes that degrade macromolecules and cell components. Many factors regulate lysosomal function and subsequent protein degradation, including luminal pH and proteolytic enzyme activity, with lysosomal dysfunction leading to profound cellular consequences. Neurons are especially vulnerable to deficiencies in autophagy substrate clearance because, without the aid of cell division, they are largely dependent on autophagy to prevent the accumulation of cellular protein and damaged organelles. An emerging body of literature from cell biology, genetics, and genome-wide association studies (GWAS) has identified lysosomal failure as a principal cellular pathology in both PD and FTD/ALS. Here, we review data from genetic and clinical studies, as well as various models of disease. Together, these data support the concept that lysosome dysfunction is a fundamental cause of both PD and FTD/ALS, rather than simply reflecting association with the endpoint of disease in the form of protein aggregates. Finally, we discuss the potential for targeting the lysosome for future therapeutics for both familial and sporadic cases of PD and FTD/ALS.

Clinical and Pathological Overlap between PD and FTD/ALS

Parkinsonism is a common presentation of FTD/ALS and is reported in as many as 20% of patients [3]. Although the clinical features of parkinsonism seen in FTD/ALS are variable, the classic akinetic-rigid syndrome is usually present. A recent, comprehensive review of the literature suggested that, although *C9ORF72* expansions are not an important genetic cause of PD, parkinsonism is a common phenotype in *C9ORF72*-related neurodegeneration [4]. Also of note, mutations in *MAPT* on chromosome 17 encoding the protein tau were originally associated with FTD with parkinsonism linked to chromosome 17 (FTPD-17) [5], now considered part of the FTD spectrum of disorders [4]. Polymorphisms at the *MAPT* locus are associated with PD by GWAS and there is an emerging appreciation of tau neuropathology in PD at post mortem [5]. Interestingly, Richardson syndrome-like phenotype is characterised by asymmetric rigid-akinetic parkinsonism and supranuclear gaze palsy, and is most commonly observed in patients with FTD with mutations in *MAPT* [6]. It is also notable that parkinsonism in FTD/ALS, such as that seen with the mutations *N267S* and *A315E* in the TDP-43-encoding gene, *TARDBP* [6], can often be alleviated with levodopa, suggesting a role of striatal DA in this phenotype.

The frequency of parkinsonism and differences in the cognitive and behavioural features of parkinsonism in clinical subtypes of FTD were recently evaluated in a nationwide hospital-based cohort established for the Clinical Research Centre for Dementia of South Korea (CREDOS)-FTD study [7]. Here, 40% of patients, of all FTD subtypes, presented with parkinsonism. Behavioural disturbances were significantly more pronounced in patients with parkinsonism than in those without. It has also been suggested that this worsening of

behavioural symptoms in patients with FTD is a marker of progression of the pathology to subcortical structures involving basal ganglia. Indeed, imaging studies with functional magnetic resonance imaging (fMRI) and DA transporter single-photon emission computed tomography (DaT-SPECT) have demonstrated a positive correlation between nigrostriatal dopaminergic function and performance on tests in executive functioning and memory [8]. Although the pathomechanism of parkinsonism in FTD is unknown, degeneration of striatal neurons and other subcortical structures has been demonstrated in pathological studies [9] (Figure 1).

Parkinson's Disease and Frontotemporal Dementia Genes Converge at the Lysosome

Although age remains the greatest risk factor for developing sporadic PD, mutations in several genes have been found to cause autosomal dominant and autosomal recessive monogenic forms of PD, and many genetic variants that modulate the risk of idiopathic disease have been identified. Many of the PD-associated genes identified by familial Mendelian inheritance patterns, such as Leucine-rich repeat kinase 2 (*LRRK2*), *ATP13A2*, and *SCNA*, encode proteins that have been shown to have a role in lysosome homeostasis and function (summarised in Table 1). A PD GWAS study in 2017 identified 17 new risk loci for PD, in addition to 24 that were previously described, and found candidate genes from associated regions to be enriched for roles in autophagic and lysosomal function [10]. Most recently, 30 additional new risk loci for PD have been identified in the largest GWAS for PD to date [11]. Examples of new candidate genes include *CLCN3*, which encodes the H⁺/Cl⁻ lysosomal pump and causes a rare juvenile neuronal ceroid lipofuscinoses (NCL) when mutated [12] and *MBNL2*, an RNA-binding protein that has been shown to be functional in mTOR signalling [13]. Furthermore, a significant burden of rare lysosomal disorder gene variants associated with PD risk was recently discovered [14]. Interestingly, mutations in *SMPD1*, which encodes the lysosomal enzyme sphingomyelin phosphodiesterase 1, are increased in patients with PD [15].

It is known that a large proportion of FTD/ALS-causing genes are related, either directly or indirectly, to pathways that maintain protein homeostasis via lysosomes and autophagy [16]. Recent findings showing that the C9ORF72 protein localises to lysosomes and regulates the lysosomal/autophagy pathway through mTORC1 signalling adds to its importance in FTD/ALS pathology and expands its enigmatic role as a major cause of both FTD and ALS [17,18]. Several other familial FTD/ALS genes, such as *GRN*, *TBK1*, *OPTN*, *CHMP2B*, *SQSTM1/p62*, *VCP/p97*, and the risk factor *TMEM106B* centre around the lysosome and protein degradation, converging pathways with high relevance in neurodegeneration [19]. Many genes associated with FTD/ALS, as well as PD, are also implicated in lysosomal storage disorders, such as *ATP13A2*, glucocerebrosidase (GCase; *GBA*), *SMPD1*, *GRN*, and *GALC* [12,20]. Furthermore, a recent report highlighted a childhood-onset neurodegenerative disorder occurring due to biallelic *SQSTM1/p62* loss-of-function variants [21], expanding the *SQSTM1/p62* phenotypic spectrum. Furthermore, among its many functions, *SQSTM1/p62* has a central role in target specification for selective autophagy and

in the ubiquitin-proteasome system (UPS) [43], identifying selective autophagy pathways as central to neurodegeneration.

Using a recently developed Weighted Protein–Protein Interaction Network Analysis pipeline as a means to define affected biological processes, risk pathways have been identified in PD as well parkinsonian syndromes, including FTD [22]. In that report, waste disposal pathways were enriched in both PD and FTD, with FTD characterised by semantic classes, such as the ‘ubiquitin proteasome system’, whereas PD was characterised by ‘autophagy’. Interestingly, familial forms of FTD/ALS with parkinsonism are most associated with genes implicated in lysosomal function, such as *GRN*, *C9ORF72*, *CHMP2B*, *VCP* [6], and *CHCHD2/10* [23], providing additional evidence for lysosomal dysfunction as a possible link between these diseases.

PD and FTD/ALS Mutations Perturb Lysosomal Biology

Autophagic Flux and Protein Aggregation

The presence of protein aggregates as the pathological hallmark of PD and FTD/ALS suggests early dysfunction of lysosomal degradation pathways and enzymes in both disorders. For example, increased cathepsin D immunoreactivity has been reported to precede the deposition of pathologic α -synuclein in post-mortem nigral neurons from patients with PD [24], implicating lysosomal perturbations before α -synuclein pathology is apparent. Previous evidence also suggests that progranulin, encoded by *GRN*, functions as a chaperone for cathepsin D in the lysosome, and that the two proteins have overall neurotrophic effects [25]. Finally, multiple autosomal recessive mutations in *CTSD*, which encodes the cathepsin D enzyme, have been identified in patients with neuronal ceroid lipofuscinoses (NCL) type 10, an aggressive neurodegenerative lysosomal storage disorder, connecting lysosomal dyshomeostasis with neurodegeneration [26,27].

Genetic mutations associated with PD or FTD/ALS have been shown to be deleterious to autophagic flux and increase the formation of protein aggregates. For example, the expression of mutant *GBA*, the most common genetic risk factor for PD, promotes α -synuclein pathology and accumulation of glucosylsphingosine (GlcSph) in transgenic mice expressing both mutant human *GBA* and human *SCNA* [28]. This is consistent with GlcSph being the toxic lipid species in mutant *GBA*-associated PD through initiation of pathological α -synuclein aggregation. Interestingly, a cofactor required for the import of glucocerebrosidase into the lysosome, LIMP2, has been linked by GWAS to the risk of developing PD [29] and was also found increased in iPSC-derived dopaminergic neurons from patients carrying the *GBA-N370S* mutation [30].

In another example, the protein encoded by *ATP13A2*, identified as the causative gene at the *PARK9* locus [31], has been proposed to participate in protein degradation in lysosomes. *Atp13a2* conditional knockout mice exhibit typical neuropathological phenotypes of the lysosomal storage disorder, NCL. Furthermore, an accumulation of the subunit c of mitochondrial ATP synthase was observed, accompanied by a delay in cathepsin D maturation [32]. Collectively, these data, coupled with the observation of increased

mitochondrial damage and oxidative stress with *ATP13A2* loss [33], suggest a role of *ATP13A2* in lysosomal function and clearance of damaged mitochondria.

Recent evidence has demonstrated protein accumulation in disease models of FTD/ALS. Knock down of *C9orf72* inhibits autophagy induction and leads to p62 accumulation [34,35], as well as aggregation of TDP-43 [35]. iPSC-derived neurons from patients with *C9ORF72* mutations were observed to have impaired basal autophagy [34,36], as well as increased sensitivity to autophagy inhibition [37]. C9ORF72, SQSTM1/p62, and FUS may also underpin a link to stress granule autophagy via a recognition cascade that, when defective, may lead to characteristic cytoplasmic protein inclusions seen in FTD/ALS [38]. It has been demonstrated that α -synuclein fibrils recruit TANK-binding kinase 1 and optineurin, encoded by *TBK1* and *OPTN*, respectively, to lysosomal damage sites and induce autophagy in microglial cells [39]. Interestingly, both proteins are also implicated in FTD/ALS, thus highlighting common pathways between PD and FTD/ALS.

Another example of altered autophagic flux and protein aggregation can be seen with toxic gain-of-function mutations in *PFN1*. Mutations in *PFN1*, which encodes the actin monomer-binding protein, profilin 1, have been shown to cause ALS [40] and toxic gain-of-function mutations in *PFN1* spur a cascade of misfolding, oligomerisation, and aggregation of mutant PFN1, leading to the formation of cytoplasmic aggregates of sequestered TDP-43 positive for p62 and ubiquitin [41,42]. Furthermore, lysosomal inhibition increases these cytoplasmic aggregates, highlighting the crucial role of the lysosome in clearing protein aggregates [42].

Besides lysosomal dysfunction, disruption of protein clearance may also occur due to the disruption of cargo-specific delivery to the lysosome. As mentioned earlier, SQSTM1/p62 has a central role in target specification for selective autophagy and in UPS [43]. Mutations in *SQSTM1/p62* can be causal for both FTD and ALS, and its presence in protein inclusions is a pathognomonic signature in the disease spectrum [44]. Interestingly, there is growing evidence that SQSTM1/p62 has a key role in the clearance of insoluble MAPT/Tau protein species [45].

Although the precise role of SQSTM1/p62 in FTD/ALS pathology is unclear, its mechanism appears to be centred around protein degradation and organelle quality control, along with another known FTD/ALS-causing gene, the AAA-ATPase known as p97 or valosin-containing protein (*VCP*) [46]. Distinct knockout and mutant *Vcp/p97* phenotypes include autophagic dysfunction and the mislocalisation of TDP-43 to the cytosol [47]. Recent findings have suggested that VCP/p97 has an important role in autophagosome maturation [48], mitochondrial quality control through UBXD1-mediated Parkin-dependent mitophagy [49,50], and the clearance of damaged lysosomes by the recruitment of the transient cofactors UBXD1, YOD1, and PLAA [51].

Overall, it is evident that the lysosome has a crucial role in both PD and FTD/ALS. Various mechanisms underlying such dysfunction have been proposed with alterations in lysosomal pH, the importance of Rabs, and lysosomal dysfunction beyond macroautophagy forming emerging themes linking PD and FTD/ALS, as discussed here (summarised in Figure 2).

Lysosomal pH

Lysosomal function depends highly on maintaining an acidic luminal pH, with shifts in pH severely disrupting degradative enzyme activity. The PD-associated proteins LRRK2, TMEM175, and ATP13A2 have all been implicated in lysosomal pH regulation. The expression of the *LRRK2-G2019S* mutation has been shown to alter lysosomal pH and lysosomal protein degradation in a kinase-dependent manner in primary astrocytes [52]. LRRK2 was recently described to directly interact with the $\alpha 1$ subunit of the vacuolar-type ATPase H^+ pump, which maintains lysosomal pH [53]. Expression of the *LRRK2-R1441C* mutation completely abolishes this interaction, leading to the loss of lysosomal acidity, decreased autolysosome maturation, and protein degradation. Interestingly, the gene encoding the vATPase $\alpha 1$ subunit, *ATP6V0A1*, lies within a PD GWAS locus, further emphasising the genetic link between lysosomal pH and PD [10].

TMEM175 is a late endosome and lysosomal K^+ channel, and its deficiency in rat primary neurons results in an instability in lysosomal pH, decreased lysosomal catalytic activity, and impaired autophagosome clearance [54]. Furthermore, this deficiency was observed to decrease GCase activity, highlighting an interaction between PD risk factors. Interestingly, it was recently demonstrated that ATP13A2 regulates synaptomagin-11, encoded by the PD-risk gene *SYT11*, at both the translational and post-transcriptional levels. Specifically, ATP13A2 depletion leads to the degradation of synaptomagin-11 and subsequent lysosomal deacidification and accumulation of α -synuclein [55], highlighting the importance of lysosomal pH regulation in disease mechanisms.

In FTD/ALS, cleaved granulins may have a role in lysosomal function by impairing the degradation of TDP-43 [56], which has also been shown to decrease autophagosome–lysosome fusion [57]. Indeed, it has been demonstrated that intracellular proteolysis of the progranulin protein generates stable, lysosomal granulins [58]. Furthermore, progranulin was recently implicated as a regulator of lysosomal acidification and biogenesis [59]. SNPs at the *TMEM106B* locus, encoding a transmembrane protein on the endolysosomal membrane, have been identified as risk alleles in FTD/ALS. Strikingly, the rare protective variant of *TMEM106B* (rs1990622), which encodes the *T185S* protein-coding change, reduces the risk of FTD in patients with concomitant *GRN* mutations by 50%, with *GRN* mutations being otherwise almost completely penetrant [60]. The TMEM106B protein has been shown to interact with vATPase accessory protein 2, and its deficiency leads to the downregulation of AP1 and V0 domain subunit expressions, as well as impaired lysosomal acidification [61]. As discussed earlier, the familial PD protein LRRK2 is also implicated in lysosomal pH via its interaction with vATPase subunits, highlighting convergent disease mechanisms. Furthermore, FTD/ALS causative mutations in *CHMP2B* impaired lysosomal trafficking, whereas reducing TMEM106B levels ameliorated these defects [62]. Given that *TMEM106B* polymorphisms may also be protective against FTD/ALS caused by *GRN* mutations, therapies targeting lysosomes and lysosomal trafficking represent a potentially promising approach for treating patients with FTD/ALS and/or PD.

Vesicular Trafficking and Rabs

Rab GTPases are key organisers of intracellular membrane trafficking and have been heavily implicated in a range of neurodegenerative diseases (reviewed in detail in [63]), with PD and FTD/ALS-associated Rabs being functionally implicated at the lysosome (Figure 3). For example, recent studies identified a subset of Rab GTPases, including Rab3, Rab8, Rab10, Rab35, Rab39b, and Rab7L1, as bona fide substrates of LRRK2 in cells [64-68]. Interestingly, Rab7L1 was shown to function *in vivo* in the maintenance of lysosomes in renal proximal tubule cells [69] and is also a candidate PD risk gene [69]. Rab7L1 has been shown to recruit LRRK2 to the *trans*-Golgi network [65,70], with deficiency in LRRK2 leading to a defect in *trans*-Golgi to lysosome protein trafficking [71]. *C9ORF72* haploinsufficiency has also been demonstrated to result in dysfunctional *trans*-Golgi network disturbances and impaired vesicular trafficking due to a loss of Rab7L1 interactions [36], highlighting Rabs as a common pathological mechanism shared between PD and FTD/ALS. Increased Rab8 phosphorylation by LRRK2-Y1699C has been shown to cause increased lipid droplet size, a hallmark of cellular distress, which is mediated by lysosomes [72]. Interestingly, upon lysosomal overload stress, LRRK2 is recruited to the lysosome, alongside Rab7L1, where it stabilises Rab8 and Rab10 through phosphorylation [73]. In the same study, it was also shown that the knockout of *Lrrk2* increases vacuolisation and lipofuscin autofluorescence, indicating that *Lrrk2* may protect against lysosomal enlargement and upregulates lysosomal secretion during lysosomal stress. Collectively, the data reported underscore a putative role of LRRK2, related Rab GTPases, and other effectors in the maintenance of lysosomal homeostasis. Identifying the specific mechanisms by which pathogenic LRRK2 mutations impact lysosomal dysfunction, and how this leads to neurodegeneration, will be important for future research.

Interestingly, guanine nucleotide exchange factors (GEFs), which act upstream of Rabs by facilitating GDP/GTP exchange, have been implicated in FTD/ALS. Disease-associated GEFs have also been shown to interact with PD-associated Rabs. For example, C9ORF72 was shown to act as a GEF complex for two PD-related Rab-GTPases: Rab39B [35,64] and Rab8a [35], a substrate of *LRRK2* [66], as well as Rab1A [34], which acts to initiate autophagy. The PD-linked protein TMEM230 is also required for Rab8a-mediated retromer trafficking, and PD-related *TMEM230* mutations induce autophagic dysfunction [74]. In addition, phosphoproteomic screening previously identified Rab8A as a downstream phosphorylation target of PINK1 [75].

Rab35, in addition to being a LRRK2 substrate, has been shown to initiate endosomal sorting complexes required for transport (ESCRT) protein recruitment in response to neuronal firing [76]. This is of particular interest in FTD/ALS because charged multivesicular body protein 2B (CHMP2B) is a component of ESCRT-III, with several roles in maturing endosome and lysosome membrane dynamics [62], and may assist with synaptic vesicle recycling in this interaction. Another FTD/ALS-related GEF, ALS2, interacts with Rab5 and has been shown to impair autophagosomal maturation as well as endolysosomal trafficking [77]. Interestingly, the two leading genetic causes of PD and FTD/ALS, *LRRK2* and *C9ORF72*, are the most heavily implicated genes in the regulation of Rabs. This suggests a putative role of Rabs in lysosomal dysfunction seen in these diseases and

highlights a potential therapeutic target that could benefit patients across a spectrum of neurodegenerative disorders.

Cargo-Specific Autophagy

The lysosome is the endpoint of several distinct cellular pathways, and its disruption can have profound effects beyond that of macroautophagy (Box 1). Mitophagy, a form of selective autophagy specific to recycling of mitochondria, is a key cellular pathway that may serve as a converging link between FTD/ALS and PD.

Several genes causative for both PD and FTD/ALS intersect along the mitophagy pathway such as *PINK1* and *PARKIN* with *TBK1*, *OPTN*, *SQSTM1/p62*, and, more recently, coiled-coil-helix-coiled-coil-helix domain containing 2 (*CHCHD2*) with *CHCHD10*. Under normal circumstances, TBK1 responds to PINK1/Parkin-induced mitochondrial ubiquitination by phosphorylating several downstream autophagy receptors, including OPTN [77] and p62 [78]. In FTD/ALS, mutated TBK1 has been shown to inhibit mitophagy [79], and phosphorylation of OPTN at Ser473 enhances its binding to ubiquitin chains and its clearance of damaged mitochondria [78]. Although SQSTM1/p62 primarily targets protein aggregates to degradative pathways, it also associates with, and sequesters, depolarised mitochondria but, unlike TBK1 and OPTN, SQSTM1/p62 is not required for efficient clearance of damaged mitochondria [79,80]. In addition, PINK1/Parkin mitophagy has been well characterised in PD, as examined in detail in a recent review [81]. Of particular interest is the recently observed interaction between Rab7A, TBK1, and Parkin/PINK1-mediated mitophagy, whereby Rab7A depletion leads to defective Parkin-dependent mitophagy [82]. Furthermore, Rab7A is a phosphorylation target of TBK1 at a homologous residue to the site phosphorylated by LRRK2 [83]. The major function associated with phosphorylation of Rabs by LRRK2 is altered membrane fusion rates due to increased Rab-membrane association [67]. Interestingly, similar effects were observed with phosphomimetic Rab7A [83]. Collectively, these data further strengthen the putative role of Rabs in lysosome dysfunction in FTD/ALS and PD.

Recently, studies have sought to unravel the relationship between CHCHD2 (implicated in PD [84]) and CHCHD10 (implicated in FTD/ALS) [85]. Both of these proteins directly interact to form heterodimers in response to mitochondrial stress, with CHCHD10 oligomerisation being dependent on CHCHD2 expression [86]. Mutations in *CHCHD10* induce TDP-43 cytoplasmic mislocalisation and potentially alter a protective nuclear transcription feedback loop comprising the mitochondrial targeted nuclear genes *COX4.2*, *NDUFB3*, and *NDUFB6* [87]. Furthermore, complexes of CHCHD2 and CHCHD10 may have a role in efficient mitochondrial respiration and support, and it has been suggested that the stabilisation of CHCHD2 in response to mitochondrial depolarisation serves as a quality control pathway, in an independent pathway from PINK1/Parkin [86]. While the precise role of these proteins is still unknown, their function appears to be related to mitochondrial membrane potential, similar to mechanisms governing PINK1/Parkin mitophagy, and further implicates mitochondrial turnover as a central mechanism in PD and FTD/ALS disease processes. However, the focus should not be placed entirely on dysfunctional mitochondria,

but rather on inoperative degradative pathways causing inadequate delivery of cargo or incomplete catabolism of damaged organelles at the lysosome.

Lysosomes Provide a Novel Target for Potential Therapeutics for PD and FTD/ALS

Previous reports have discussed targeting autophagy [88,89] and mitophagy induction [81] in the treatment of PD. However, the evidence implicating lysosomal dysfunction across PD and FTD/ALS presented in this review provide a compelling rationale for targeting the lysosome itself. Simply increasing the induction of autophagy might increase lysosomal stress and could, in fact, be deleterious by overloading the protein degradation system. Surprisingly, a recent study that used correlative light electron microscopy definitively showed that Lewy bodies in fact mainly comprise crowded lysosomes, autophagosomes, and lipids [90]. Furthermore, idiopathic PD brains have an accumulation of lysosomal proteins in the basal ganglia at post mortem [91], suggesting that lysosomal dysfunction has a role in not only familial PD, but also idiopathic and sporadic PD. For instance, it was shown that treatment with the lysosomal enzyme α -iduronate-2-sulfatase reduced oxidised DA levels in iPSC-dopaminergic neurons from patients with familial PD and idiopathic PD [92].

One potential approach in targeting the lysosome would be the use of small molecules capable of modulating lysosomal properties, such as pH. Sandor *et al.* used a novel approach that combined the use of transcriptomics and Connectivity Map data to identify candidate drugs capable of ameliorating genetic perturbations associated with *LRRK2* mutations [93]. One small molecule identified was the zinc/copper ionophore, clioquinol. This molecule was then shown to increase autolysosome maturation and lysosome acidity in neurons expressing the *LRRK2-R1441C* mutation, via the stabilisation of the v-ATPase proton pump [53]. Such approaches have potential for identifying candidate drugs for repurposing as a lysosome target in both PD and FTD/ALS. Another potential target for small molecules is the modulation of lysosomal calcium levels and local calcium release, which are altered in various models of both PD and FTD/ALS [53,94]. For example, the TRPML1 agonist, MLSA1, has been shown to rescue lysosomal dysfunction *in vitro* [95]. In addition, the small-molecule modulator of GCase, NCGC00188758, enhances the clearance of α -synuclein in iPSC-dopaminergic neurons from patients harbouring *SCNA*, *GBA*, or *ATP13A2* mutations, as well as from patients with idiopathic PD [96]. Given that mutations in both *GBA* and *LRRK2* are risk factors for sporadic PD, and that shared molecular mechanisms underlying sporadic and familial PD have been identified, targeting these two proteins may ameliorate lysosomal deficits present in most patients with PD.

However, although pharmacological strategies might be beneficial, issues such as blood–brain barrier penetration and lack of specificity may hinder their application in the clinic. The use of nanoparticles to deliver therapeutics is an attractive concept, because this approach can deliver drugs to specific tissues and provide controlled-release therapy. Poly(DL-lactide-*co*-glycolide) acid nanoparticles have been shown to restore impaired lysosomal function in several genetic *in vitro* models of PD [97]. Whether such nanotechnology-based strategies can be used in an *in vivo* setting remains to be determined.

Gene therapy also provides a potential means of improving current therapy for these diseases. While nondisease-modifying treatments to alleviate PD symptoms have shown promising results, no trial implementing disease-modifying therapies is yet to meet its primary endpoint [98]. Nonetheless, such approaches are attractive and demonstrate potential. For example, progranulin gene therapy was recently shown to improve lysosomal function in a mouse model of FTD, thereby supporting protein-boosting therapies for FTD [99].

One consideration of targeting the lysosome for therapeutic purposes in these diseases is the potential deleterious consequences of boosting lysosomal function across the body. Several core functions of the lysosome, including its ability to be exocytosed and to scavenge nutrients from proteolytic degradation, become enhanced in cancer cells and contribute to cancer progression [100]. Such malignant adverse effects would need to be taken into consideration for potential future lysosome-targeting therapies to be beneficial.

Concluding Remarks

Patients with PD and FTD/ALS, and models of these diseases, display accumulation of lysosome-related proteins and alterations in protein clearance and lysosome homeostasis. Therefore, it is unsurprising that these two diseases were recently described as *forms frustes* of lysosome storage diseases [101]. The recent evidence discussed in this review has increased our understanding of how pathogenic PD and FTD/ALS mutations disrupt lysosome function.

The evidence discussed herein also suggests that protein aggregates present in disease are primarily an outcome of upstream disease mechanisms, rather than being pathogenic themselves. The Bradford Hill criteria for causation defines nine criteria to provide epidemiological evidence of a causal relationship between a presumed cause and an observed effect [102]. These criteria were recently applied to records from 16 different studies on patients with PD, and only two of the nine criteria were satisfied [103]. These results indicate that human studies of protein aggregation in PD do not meet a minimum set of criteria to support a causal role of protein aggregates in disease. If protein aggregates are not directly causal, they may still contribute to neurodegeneration by accelerating disease, be epiphenomenon, or be protective by guarding against toxic soluble forms of proteins by sequestering them as insoluble forms. These frameworks are difficult to resolve because of the inability to probe brain tissue in real time and examine the potentially dynamic components in cellular proteostatic mechanisms and, therefore, remain a heavily debated topic in the field.

Furthermore, if lysosomal dysfunction is indeed a convergent mechanism between PD and FTD/ALS, why do some patients develop PD and others FTD/ALS? One possibility is the idea of selective neuronal and regional vulnerability. Selective neuronal vulnerability refers to the fact that the pathology associated with the disease only affects particular neurons in particular regions of the brain, a characteristic of all neurodegenerative diseases. Critical and intrinsic pathways that are close to failure in the normal brain and that fail in the diseased brain have been suggested, including the endolysosomal system [104]. Selective neuronal

vulnerability in different diseases may also involve neuron-specific combinations of dysfunction in other pathways, such as calcium signalling and mitochondria, aggravated by advancing age, gene predisposition, and environmental factors.

Although lysosome disruption has been demonstrated in a large number of familial mutations, as well as sporadic cases, PD and FTD/ALS are heterogeneous diseases. Parkinsonism is not a universal symptom seen in all patients with FTD/ALS, suggesting that there are diverging mechanisms between different subtypes of the disease, including involvement of the lysosome (see Outstanding Questions). Establishing which patients would benefit from early interventions targeting the lysosome, in a precision medicine approach, should be a consideration for future research. For example, the development of biomarkers to identify patients with underlying lysosomal disruption would help guide the implementation of future therapeutics targeting the lysosome. However, given the large number of cellular pathways that are regulated by, and converge on, the lysosome, it is possible that such therapeutics could benefit a large number of patients and have wider applicability.

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Glossary

Chromosome 9 open reading frame 72 (C9ORF72)

hexanucleotide GGGGCC repeat expansion in *C9ORF72* was the first identified genetic link between FTD and ALS and is the most common mutation associated with familial FTD/ALS. The protein appears to function in the DNA damage response and autophagy, and functions as a GEF for GTPases.

Dopamine (DA)

neurotransmitter derived from tyrosine. It is released by dopaminergic neurons of the midbrain onto medium spiny neurons in the striatum (among other targets) to help regulate movement.

Fused in sarcoma (FUS)

a DNA/RNA-binding protein with a role in transcription regulation, RNA splicing, RNA transport, DNA repair, and the DNA damage response. Mutations in *FUS* have been linked to ALS, and *FUS* is a pathological hallmark in subgroups of patients with FTD with inclusions immunoreactive for both *FUS* and ubiquitin.

Glucocerebrosidase (GCase; GBA)

an enzyme that cleaves glucocerebroside, an intermediate protein in glycolipid metabolism. It is encoded by *GBA*. Patients homozygous for mutations in *GBA* have Gaucher's disease, a lysosomal storage disease. A subset of patients with Gaucher's disease exhibit

parkinsonian symptoms. Individuals who carry a heterozygous *GBA* mutation have an increased risk of developing PD.

Induced pluripotent stem cells (iPSCs)

pluripotent cells derived from differentiated adult cells that can be manipulated to produce different cell types, including neurons.

Leucine-rich repeat kinase 2 (LRRK2)

a large multidomain protein with GTPase and kinase domains that has multiple intracellular roles, including the regulation of autophagy. It is encoded by *LRRK2*. Mutations in *LRRK2* cause familial PD, and polymorphisms in *LRRK2* confer a risk of sporadic PD.

Substantia nigra pars compacta (SNc)

literally ‘black substance’, a region of the midbrain where pigmented neuromelanin-containing dopaminergic neurons are located. It is divided into two main parts: the pars compacta (SNc) and pars reticulata (SNr). Loss of dopaminergic neurons from the SNc is the pathological hallmark of PD. Lewy bodies and neurites are present in some of the remaining neurons in most cases of PD.

α -Synuclein (SNCA)

presynaptically enriched protein that acts in conjunction with SNARE proteins to regulate neurotransmitter release. It is the major component of two pathological hallmarks of PD: proteinaceous aggregates, termed Lewy bodies and Lewy neurites. The α -synuclein protein is encoded by *SNCA*.

Tar DNA-binding protein 43 (TDP-43)

a predominately nuclear-localised protein that binds to both DNA and RNA, with multiple functions in pre-mRNA splicing and translational regulation. A hyperphosphorylated and cleaved form of TDP-43 that mislocalises to the cytosol is a major pathological hallmark of FTD/ALS. TDP-43 is encoded by *TARDBP*.

Tau

a microtubule-associated protein that is present in the axons of neurons, and helps to regulate axonal transport. Tau protein is encoded by *MAPT*.

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Highlights

PD and FTD-associated genetic variants are heavily implicated in lysosome function; lysosomal disorder gene variants are associated with increased PD risk, and waste disposal pathways are enriched in both PD and FTD in weighted protein–protein interaction network analyses.

Several PD and FTD-associated genes have now been implicated in the regulation of lysosomal pH, a potential underlying mechanism of lysosome dysfunction seen in PD and FTD.

LRRK2 and *C9ORF72*, the leading genetic causes of PD and FTD, respectively, are both heavily implicated in the regulation of Rabs, suggesting a putative role of Rabs in lysosomal dysfunction in these diseases.

Small molecules targeting the lysosome, capable of altering lysosomal pH, have been shown to reverse lysosome dysfunction in models of disease, highlighting future therapeutic targets and potentials.

Box 1.**The Impact of Lysosome Dysfunction Beyond Protein Aggregation**

Protein degradation and aggregation are typically at the forefront of discussions regarding lysosomes in neurodegenerative diseases. This is perhaps not too surprising given that, for many years, the lysosome has been considered simply as a cellular waste-processing plant, where cargo delivered through autophagy pathways and the late endosomes is digested or recycled. However, lysosomes perform a range of functions implicated in a variety of cellular pathways, often performed in a cell-specific manner, and many of these cellular functions are relevant to both PD and FTD/ALS.

For example, lysosomes are known to engage in physical contact with other organelles. Contact between the lysosome and mitochondria has been shown to regulate mitochondrial fission via Rab7 GTP hydrolysis [105]. These contact sites may additionally be implicated in metabolism, because lysosome-derived metabolites are transferred into the mitochondrial matrix, helping to fuel the tricarboxylic acid cycle [100]. Given that alterations in mitochondria and lysosomes are often present concomitantly in cells from patients with neurodegenerative conditions and from animal models of disease, there may be a close functional link between these two organelles in neurodegenerative diseases.

The function of lysosomes in immune responses has also been recently highlighted, especially the contribution to macrophages in both innate and adaptive immune responses. Macrophages are phagocytic cells that ingest and digest pathogens, dead cells, and debris, and produce cytokines and chemokines to recruit other immune cells. Lysosomes, which are the destination of cells and pathogens engulfed by macrophages, have a crucial role in the processing and secretion of inflammatory signals. For instance, secretory lysosomes can secrete or degrade inflammatory cytokines to regulate the release of cytokines in the immune response [106]. In the adaptive immune system, lysosomal proteolysis generates peptides that bind major histocompatibility complex (MHC) molecules, which, in turn, allows for the presentation of information from pathogens to the T lymphocyte system [107]. Chronic inflammation is a common characteristic of both PD and FTD/ALS, with elevated levels of circulating inflammatory markers in patients with PD and FTD/ALS [108,109]. Although the involvement of lysosomes in this inflammatory component of disease remains to be elucidated, it is an intriguing area for future research.

Outstanding Questions

Is lysosomal dysfunction the mechanism underlying the presence or absence of parkinsonism in subtypes of FTD/ALS?

What makes different groups of neurons vulnerable in PD and FTD/ALS, given that lysosome dysregulation is implicated in both?

Can biomarkers be developed to identify patients who would benefit most from early interventions targeting the lysosome?

Would future clinical trials of lysosomal therapies require stratification of patients into those most vulnerable to lysosomal deficits?

Does lysosome dysfunction have other cellular consequences beyond protein degradation, such as immune function, in PD and FTD/ALS?

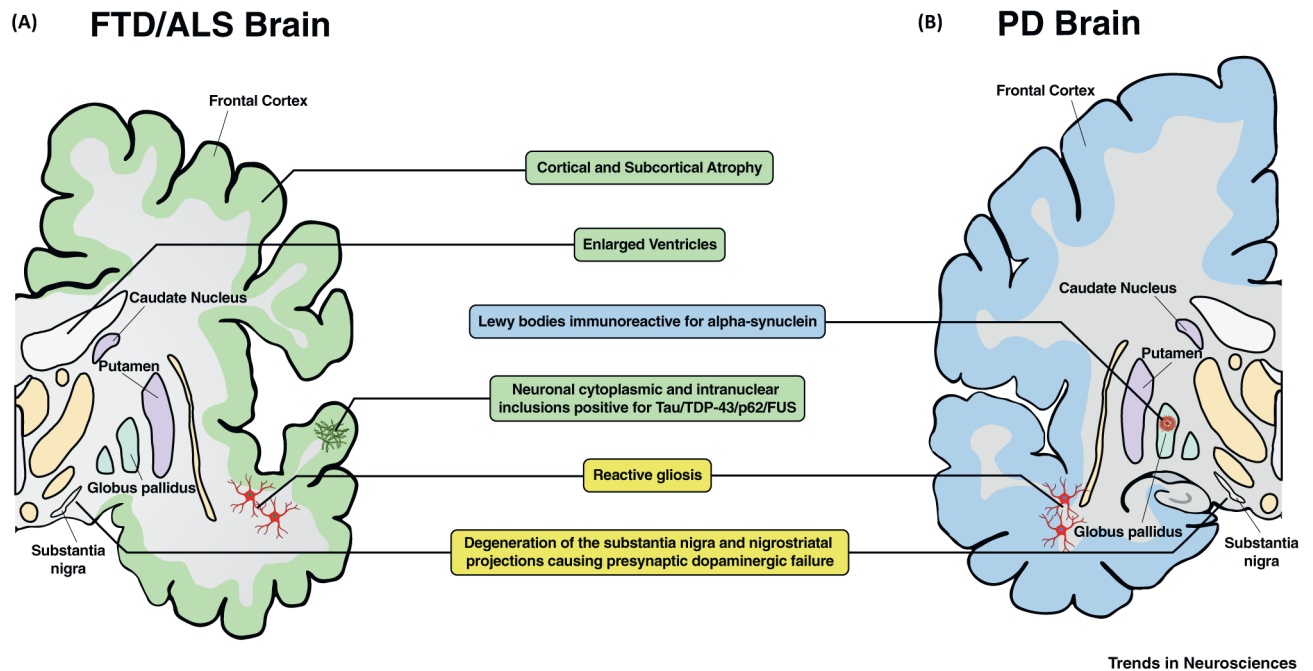
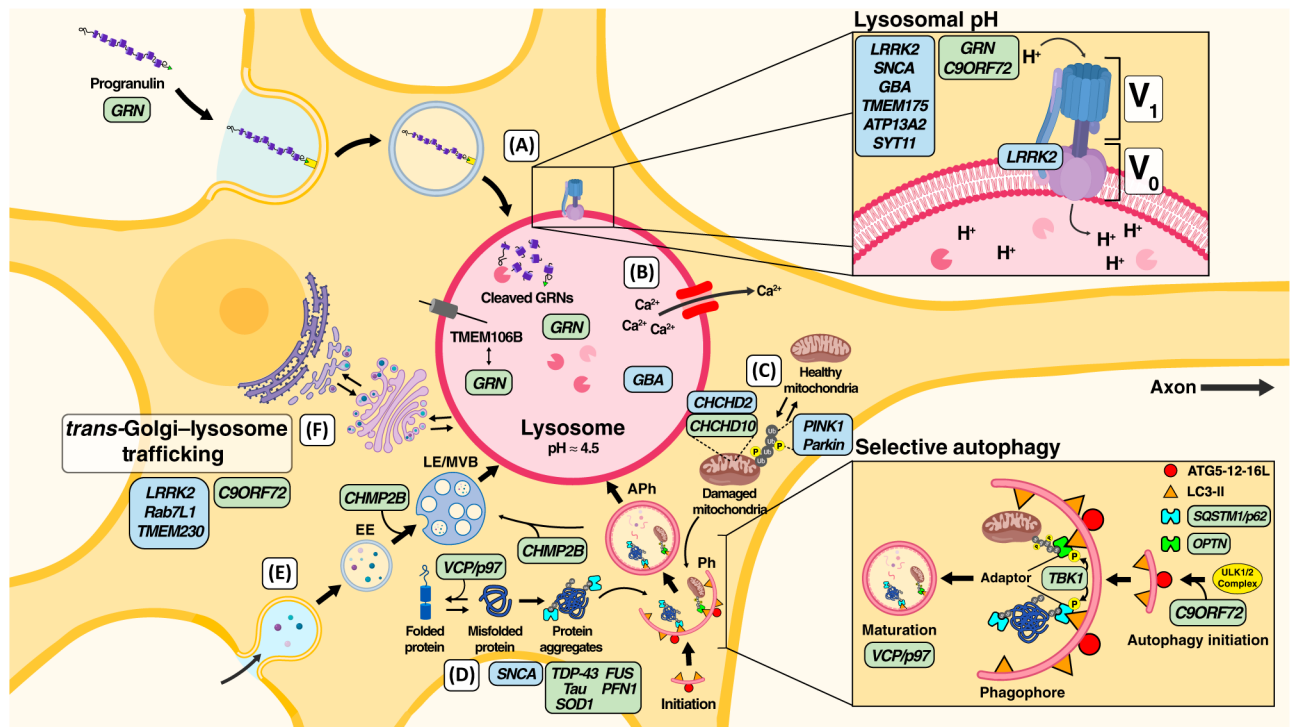


Figure 1. Frontotemporal Dementia/Amyotrophic Lateral Sclerosis (FTD/ALS) and Parkinson's disease (PD) Brain Pathology.

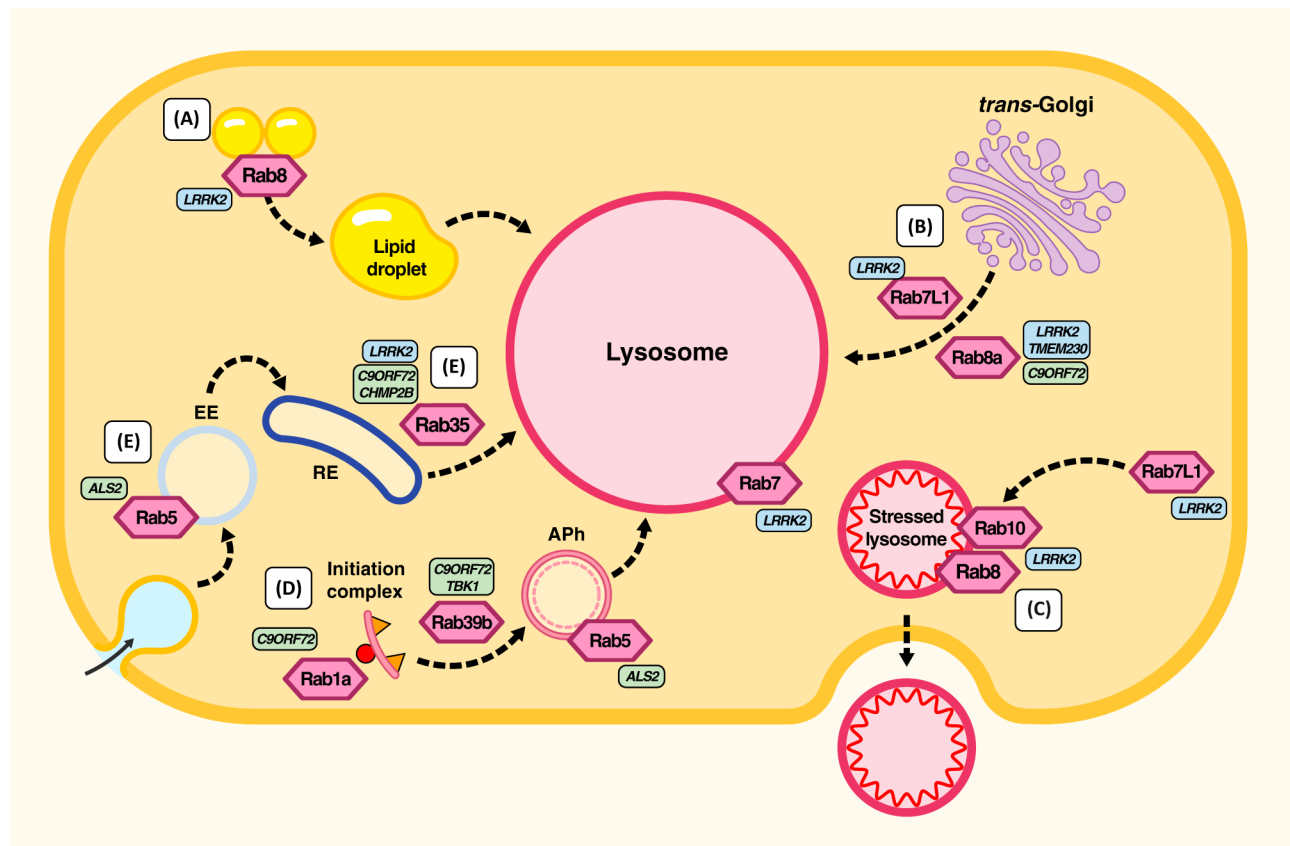
(A) FTD/ALS pathology is typically characterised by cortical and subcortical atrophy, enlarged ventricles and Tar DNA-binding protein 43 (TDP-43)/Tau/p62/Fused in sarcoma (FUS)-positive inclusions. (B) Protein aggregates in the form of Lewy bodies are the defining pathological hallmark of PD. Common pathology also exists between FTD/ALS and PD, with reactive gliosis and degeneration of the substantia nigra and nigrostriatal projections observed. Green, FTD/ALS; blue, PD; yellow, FTD/ALS and PD. Adapted from [6].



Trends in Neurosciences

Figure 2. Frontotemporal Dementia/Amyotrophic Lateral Sclerosis (FTD/ALS) and Parkinson's disease (PD)-Associated Genes Converge on the Lysosome.

Recent findings provide evidence for a role of both PD and FTD/ALS-associated genes in multiple aspects of lysosome biology, including (A) regulation of lysosomal pH, (B) lysosomal calcium release, (C) mitophagy and selective autophagy, (D) protein quality control and macroautophagy, (E) the endolysosomal pathway, and (F) *trans*-Golgi-lysosome trafficking. Blue boxes, PD; green boxes, FTD/ALS. Abbreviations: APH, autophagosome; C9ORF72, chromosome 9 open reading frame 72; CHCHD2/10, coiled-coil-helix-coiled-coil-helix domain containing 2/10; CHMP2B, charged multivesicular body protein 2B; EE, early endosomes; FUS, fused in sarcoma; LE, late endosomes; LRRK2, Leucine-rich repeat kinase 2; MVB, multivesicular body; Ph, phagophore; SNCA, α -synuclein; TDP-43, Tar DNA-binding protein 43; VCP, vasolin-containing protein.



Trends in Neurosciences

Figure 3. Rab Proteins Have Convergent Roles in Frontotemporal Dementia/Amyotrophic Lateral Sclerosis (FTD/ALS) and Parkinson's disease (PD).

Rab proteins are heavily implicated in both FTD/ALS and PD, being shown to interact with several disease-associated proteins in a number of pathways that converge on the lysosome. (A) Leucine-rich repeat kinase 2 (LRRK2) mediates lipid droplet formation via Rab8. (B) Compartmentalisation and trafficking of lysosomal enzymes and membrane proteins in the *trans*-Golgi to the lysosome is crucial for lysosome function and is modulated by LRRK2, TMEM230 and chromosome 9 open reading frame 72 (C9ORF72) via their interaction with Rab proteins. (C) Effective lysosome exocytosis during stress-induced lysosome overload is key to cell survival and has been shown to be mediated by LRRK2 and its substrates, Rab7L1, Rab10, and Rab8. (D) C9ORF72 has been shown to act as a guanine exchange factor (GEF) for several Rab proteins implicated in autophagosome formation. (E) Both PD and FTD/ALS-associated proteins have been shown to modulate different stages of the endolysosomal pathway via their interactions with Rabs, intersecting at both early endosomes (EE) and recycling endosomes (RE). Blue boxes, PD; green boxes, FTD/ALS. Abbreviation: APh, autophagosome.

Table 1.

Genes and Risk Factors Associated with PD and FTD/ALS That Have Been Implicated in Lysosomal Function^a

Gene	Inheritance	Pathology	Functions implicated at lysosome
<i>SNCA (PARK1/4)</i>	AD	Early-onset, severe PD with Lewy bodies	Recruits FTD/ALS-associated proteins to lysosomal damage sites
<i>Parkin (PARK2)</i>	AR	Early-onset PD with slow progression and no Lewy bodies in most cases	Ubiquitin ligase that catalyses ubiquitin transfer to mitochondria for mitophagy
<i>PINK1 (PARK6)</i>	AR	Early-onset PD with slow progression and Lewy body pathology	Has an essential role in mitophagy
<i>LRRK2 (PARK8)</i>	AD	Typical, late-onset PD with Lewy bodies	Regulates lysosomal pH via vATPase a1 subunit; regulates lysosome to <i>trans</i> -Golgi trafficking and lysosome homeostasis maintenance via Rab phosphorylation
<i>ATP13A2 (PARK9)</i>	AR	Juvenile-onset atypical parkinsonism (Kufor–Rakeb); pathology demonstrates ceroid lipofuscinosis	Regulates lysosome pH via SYT11 and is functional in the clearance of damaged mitochondria
<i>GBA</i>	N/A	Typical, late-onset PD with Lewy bodies	GBA mutations lead to accumulation of cholesterol in lysosomes and MLBs; modulating GCase activity boosts α -synuclein activity
<i>TMEM175</i>	N/A	PD; not yet described	Regulates lysosomal pH
<i>VPS35 (PARK17)</i>	AD	Typical, late-onset PD with unknown pathology	Role in endosome–Golgi complex trafficking; mutations impair autophagy
<i>TMEM230</i>	N/A	Typical, late-onset PD with unknown pathology	Mutations induce autophagic dysfunction
<i>C9ORF72</i>	AD	Adult-onset ALS/FTD with ubiquitin/p62/TDP-43/FUS proteinopathy	Regulates autophagy induction and stress granule autophagy; acts as a GEF for 2 PD-related Rab-GTPases
<i>GRN</i>	AD	FTLD-TDP with ubiquitin/p62/TDP-43 proteinopathy	Chaperones cathepsin D and regulates lysosomal pH
<i>TBK1</i>	AD	Adult-onset ALS/FTD with ubiquitin/p62/TDP-43/FUS proteinopathy	Required for efficient clearance of damaged mitochondria and is recruited to lysosomes by α -synuclein
<i>OPTN</i>	AD	Adult-onset FTD/ALS with unknown pathology	Required for efficient clearance of damaged mitochondria and is recruited to lysosomes by α -synuclein
<i>CHMP2B</i>	AD	Adult-onset FTD/ALS with TDP-43/ubiquitin proteinopathy	Regulates lysosomal trafficking and is an ESCRT-III component
<i>CHCH2/10</i>	AD	Late-onset PD or adult-onset FTD/ALS with unknown pathology	Responsible for mitochondrial quality control
<i>TMEM106B</i>	N/A	Adult-onset FTD with tau proteinopathy	Interacts with vATPase accessory protein 2 and regulates lysosomal pH
<i>VCP</i>	AD	FTD with TDP-43/ubiquitin proteinopathy	Functional in autophagosome maturation and clearance of damaged lysosomes
<i>SQSTM1</i>	AD	Adult-onset FTD/ALS with TDP-43/ubiquitin/FUS proteinopathy	Targets protein aggregates for lysosomal degradation

^a Abbreviations: AD, autosomal dominant; AR, autosomal recessive.